PAPER 3

Introduction Alzheimer’s disease is a progressive neurodegenerative disorder that affects cognitive function and memory. Early detection is crucial for effective intervention and management. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown promise in medical image analysis. This project aims to develop a CNN-based classification model to identify Alzheimer’s disease stages from MRI scans using the ResNet-18 architecture.

Research Paper Reference

- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- Wang, S., Wang, H., Cheung, A.C., et al. (2021). Alzheimer's Disease Classification using Deep Convolutional Neural Networks. Medical Image Analysis, 67, 101856.

Methodology

1. **Dataset**: The dataset consists of MRI images classified into four categories, indicating different stages of Alzheimer’s disease.
2. **Preprocessing**:
    - Images are resized to 150x150 pixels.
    - Normalization is applied to standardize pixel values.
    - Data augmentation techniques are used to enhance model generalization.
3. **Model Selection**:
    - ResNet-18 is chosen for its efficiency and strong feature extraction capability.
    - Pre-trained weights from ImageNet are used for transfer learning.
4. **Training Process**:

- Loss function: CrossEntropyLoss
- Optimizer: Adam with a learning rate of 0.001
- Number of epochs: 10
- Training and validation batches are processed on a GPU-enabled environment.

Evaluation Metrics

- **Accuracy**: Measures the overall correctness of the model's predictions.
- **Loss**: Tracks model convergence during training.
- **Precision, Recall, and F1-Score**: Evaluates classification performance for each class.
- **Confusion Matrix**: Provides insight into class-wise prediction errors.
- **ROC Curve & AUC Score**: Measures the model's discriminatory power.

Results and Analysis

- The model achieved a final training accuracy of **99.20%**.
- The classification report indicates strong performance across all classes with a weighted F1-score of **98%**.
- Loss values decreased steadily, indicating successful model convergence.
- The confusion matrix and ROC analysis confirm high classification accuracy and robustness.

Conclusion and Future Work This project demonstrates the effectiveness of deep learning in Alzheimer's disease classification. The high accuracy and generalization ability suggest that CNN-based models can assist in early detection. Future work may include:

- Incorporating larger and more diverse datasets.
- Enhancing interpretability using Grad-CAM visualization.
- Optimizing computational efficiency for real-time deployment in clinical settings.
- Exploring ensemble models for further accuracy improvements.

This project provides a solid foundation for leveraging AI in medical diagnostics, contributing to improved patient outcomes and early disease intervention.

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Epoch 1/10, Loss: 0.4397035073838197, Accuracy: 81.10%
Epoch 2/10, Loss: 0.19119447202247103, Accuracy: 92.83%
Epoch 3/10, Loss: 0.10888074577087536, Accuracy: 95.96%
Epoch 4/10, Loss: 0.06079086299832852, Accuracy: 97.96%
Epoch 5/10, Loss: 0.05147202484185982, Accuracy: 98.18%
Epoch 6/10, Loss: 0.05500557604100322, Accuracy: 98.13%
Epoch 7/10, Loss: 0.034585122778298685, Accuracy: 98.71%
Epoch 8/10, Loss: 0.026563137739924515, Accuracy: 99.05%
Epoch 9/10, Loss: 0.03240525908172458, Accuracy: 98.79%
Epoch 10/10, Loss: 0.024387853883263234, Accuracy: 99.20%
Classification Report:
               precision    recall  f1-score   support

           0       0.98      0.99      0.99       179
           1       1.00      1.00      1.00        12
           2       0.98      0.98      0.98       640
           3       0.98      0.97      0.97       448

    accuracy                           0.98      1279
   macro avg       0.98      0.99      0.99      1279
weighted avg       0.98      0.98      0.98      1279
```

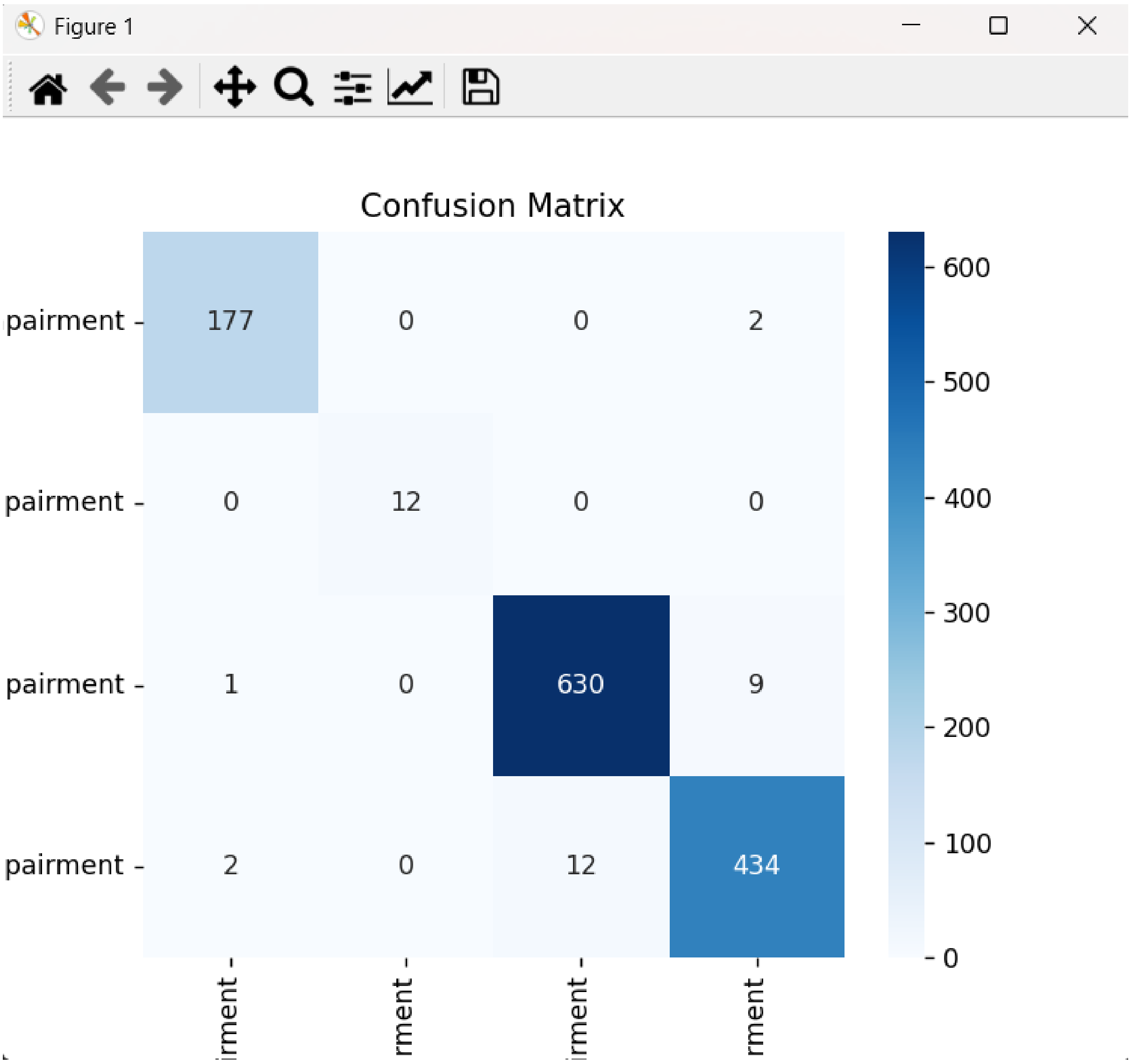
Figure 1
Confusion Matrix
pairment
pairment
pairment
pairment
177
0
0
2
0
12
0
0
1
0
630
9
2
0
12
434
600
500
400
300
200
100
0
irment
rment
irment
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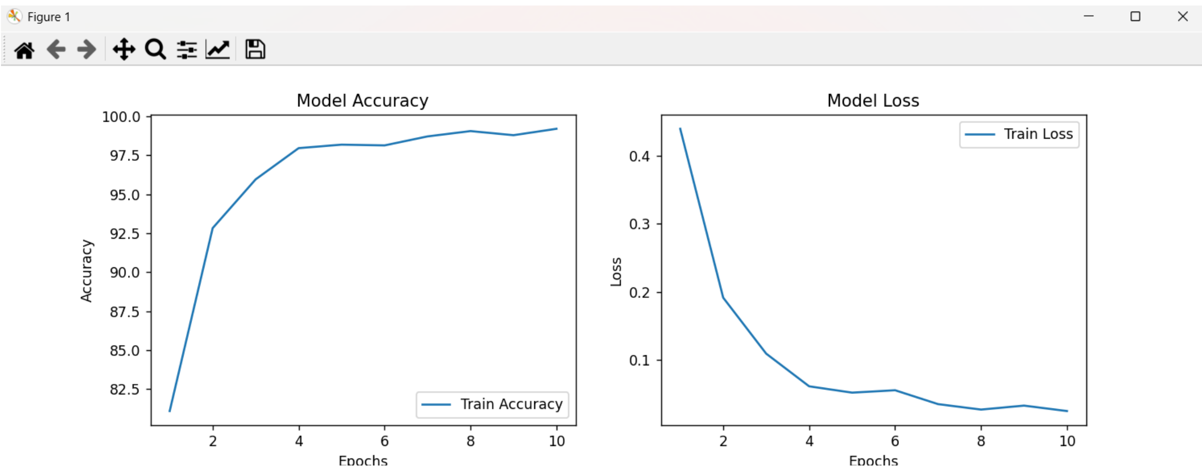
Figure 1
Model Accuracy
Accuracy
Epochs
Train Accuracy
Model Loss
Loss
Epochs
Train Loss

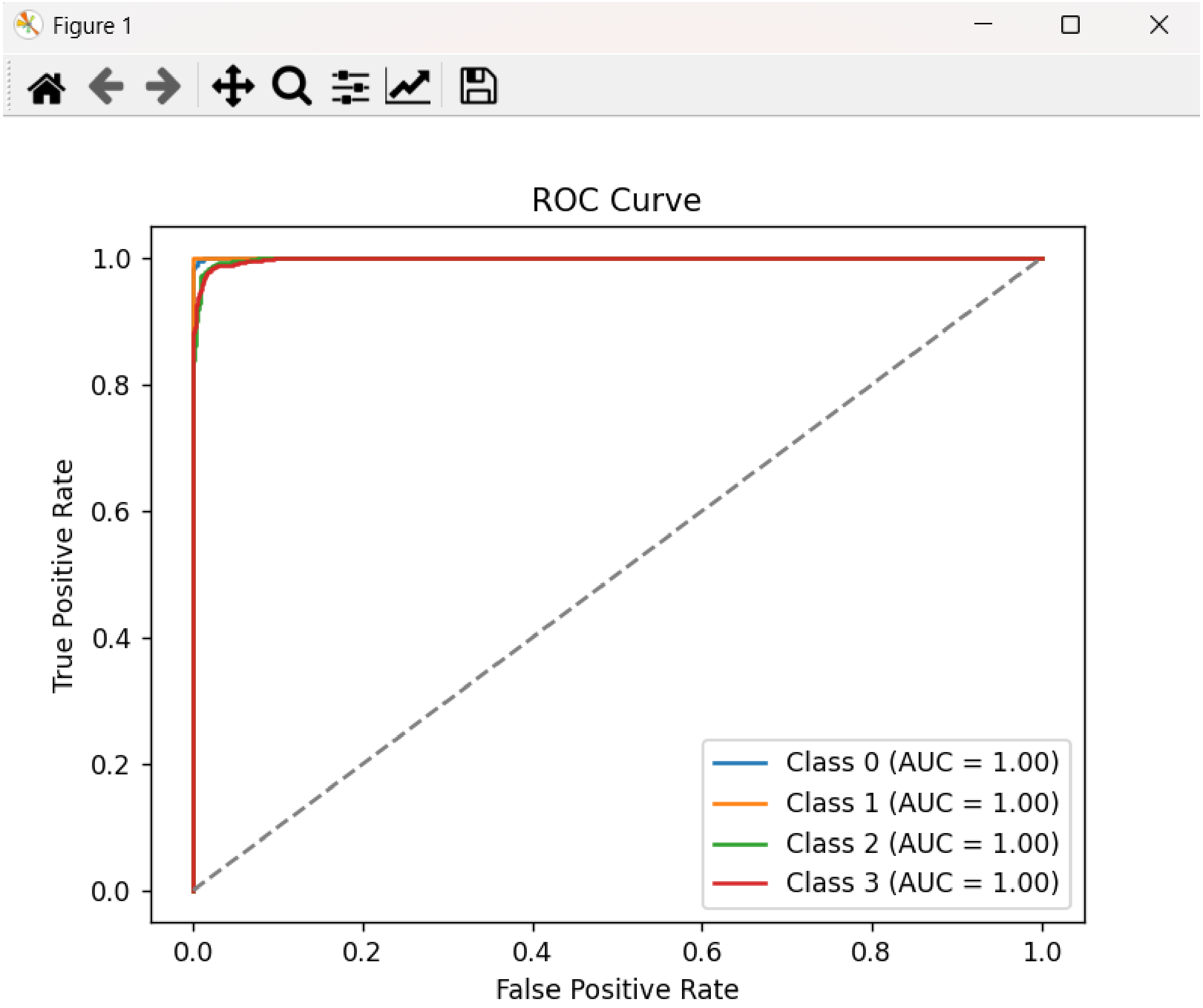
Figure 1
ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Class 0 (AUC = 1.00)
Class 1 (AUC = 1.00)
Class 2 (AUC = 1.00)
Class 3 (AUC = 1.00)